In May 2005, Kittilä and Ivalo flooded together.

The snow had warned us two months earlier.

The 1 March snow flagged 2005 in advance. Then both rivers flooded in the same spring.



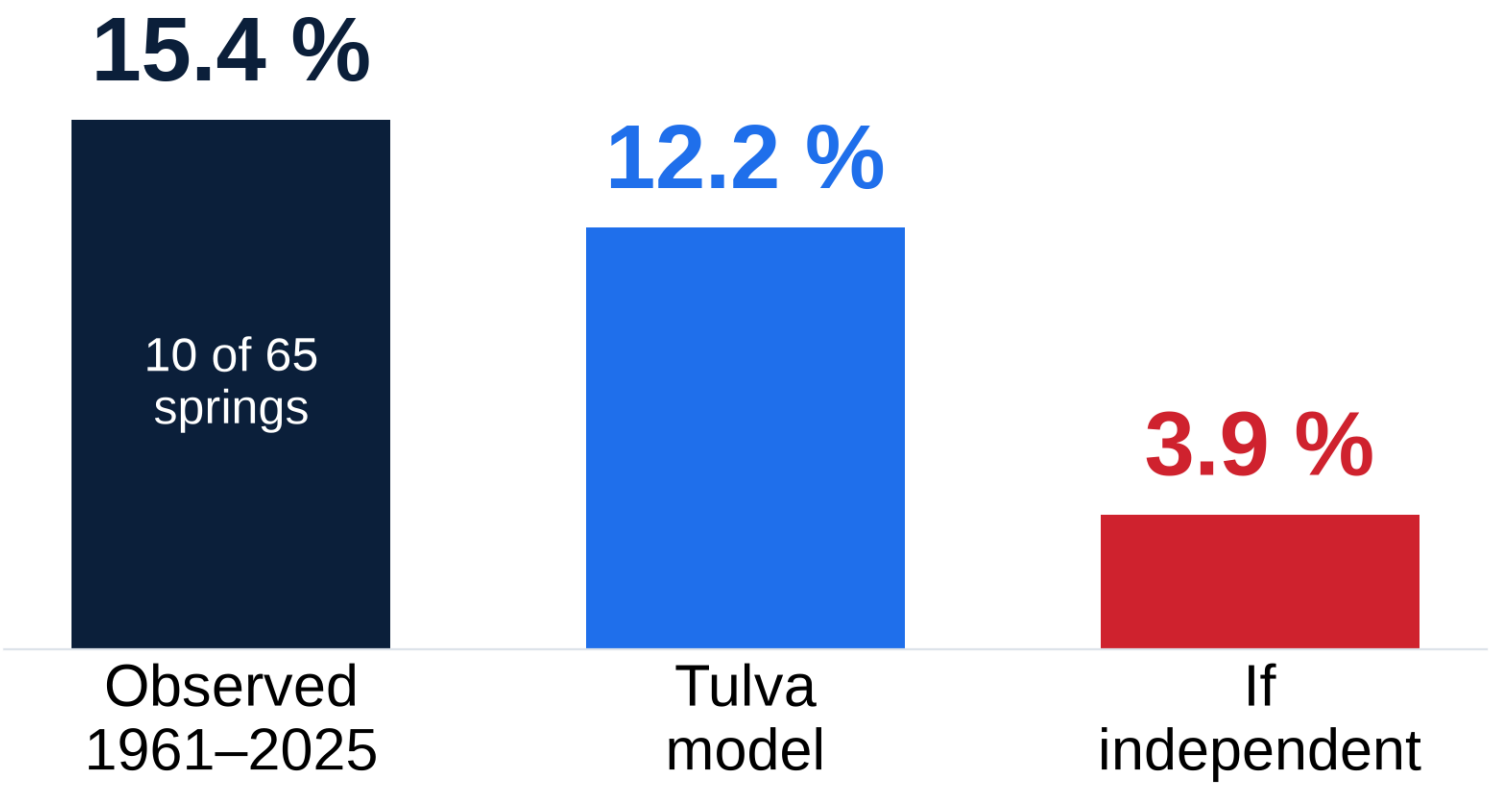
Tulva app: spring 2005 replay

Source: SYKE discharge 1961–2025 and 1 March snow (SWE); Tulva back-test (out/backtest.json)

THE GAP

Finland's rivers flood together, not one at a time

How often do ≥4 of 14 river areas flood in the same spring?



Springs with ≥4 of 14 areas over their 1-in-10 level

15.4 %
observed (10 of 65 years)

12.2 %
Tulva joint model

3.9 %
if areas were independent

Reinsurance renews 1 January, before the snow signal arrives.

Source: SYKE 1961–2025 (+ SMHI Torne), Tulva model (exact); vesi.fi / OP (1-in-50 rule)

THE PRODUCT

Tulva Index: bought in winter, pays in spring

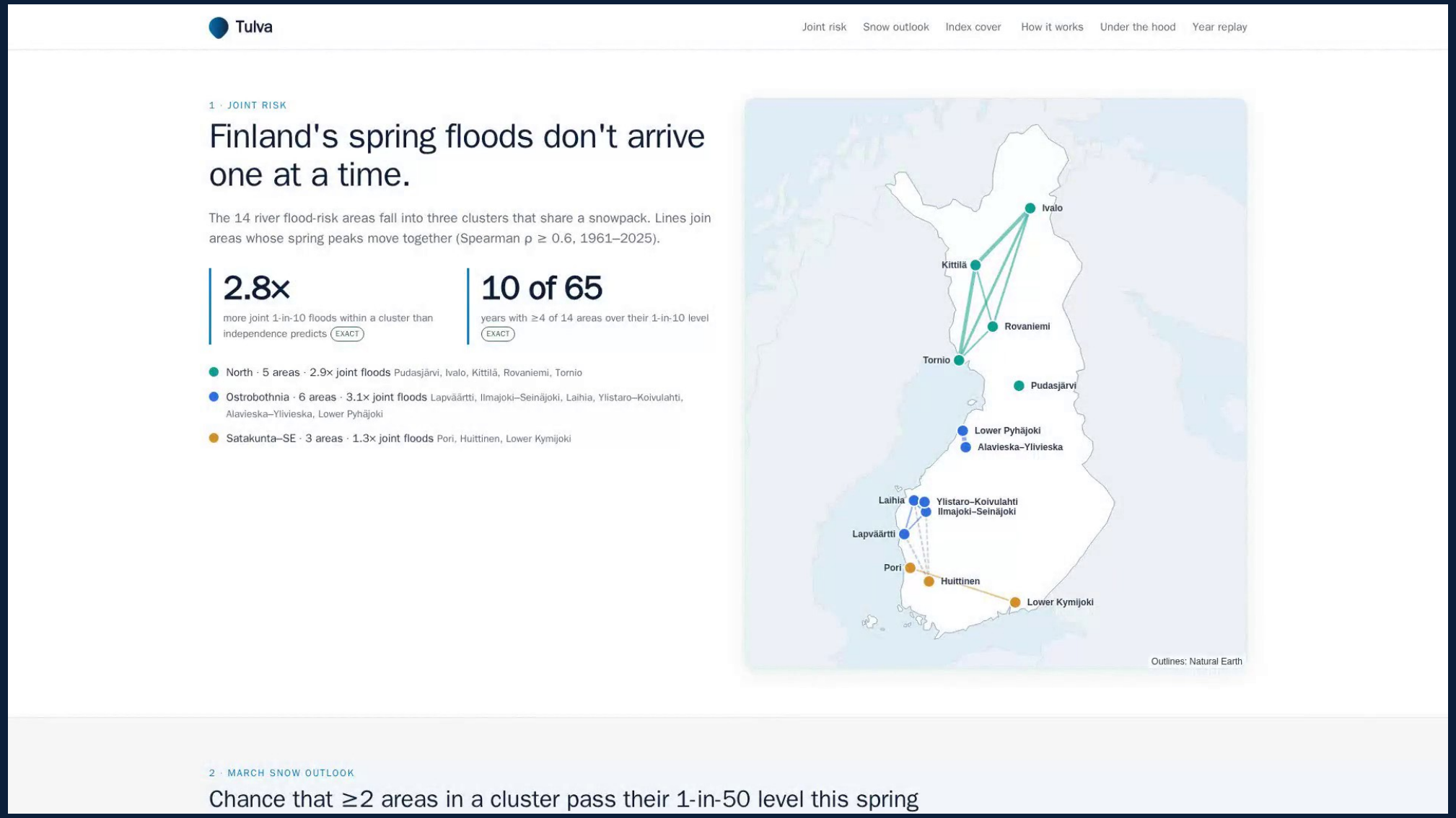
Pays €10M if ≥2 of 5 northern gauges pass their 1-in-50 level this spring

MARKET PRICE PER YEAR



Today's cat models assume how areas flood together. **Tulva measures it from 65 years of river data, ties it to this winter's snow, and prices it as a cover.**

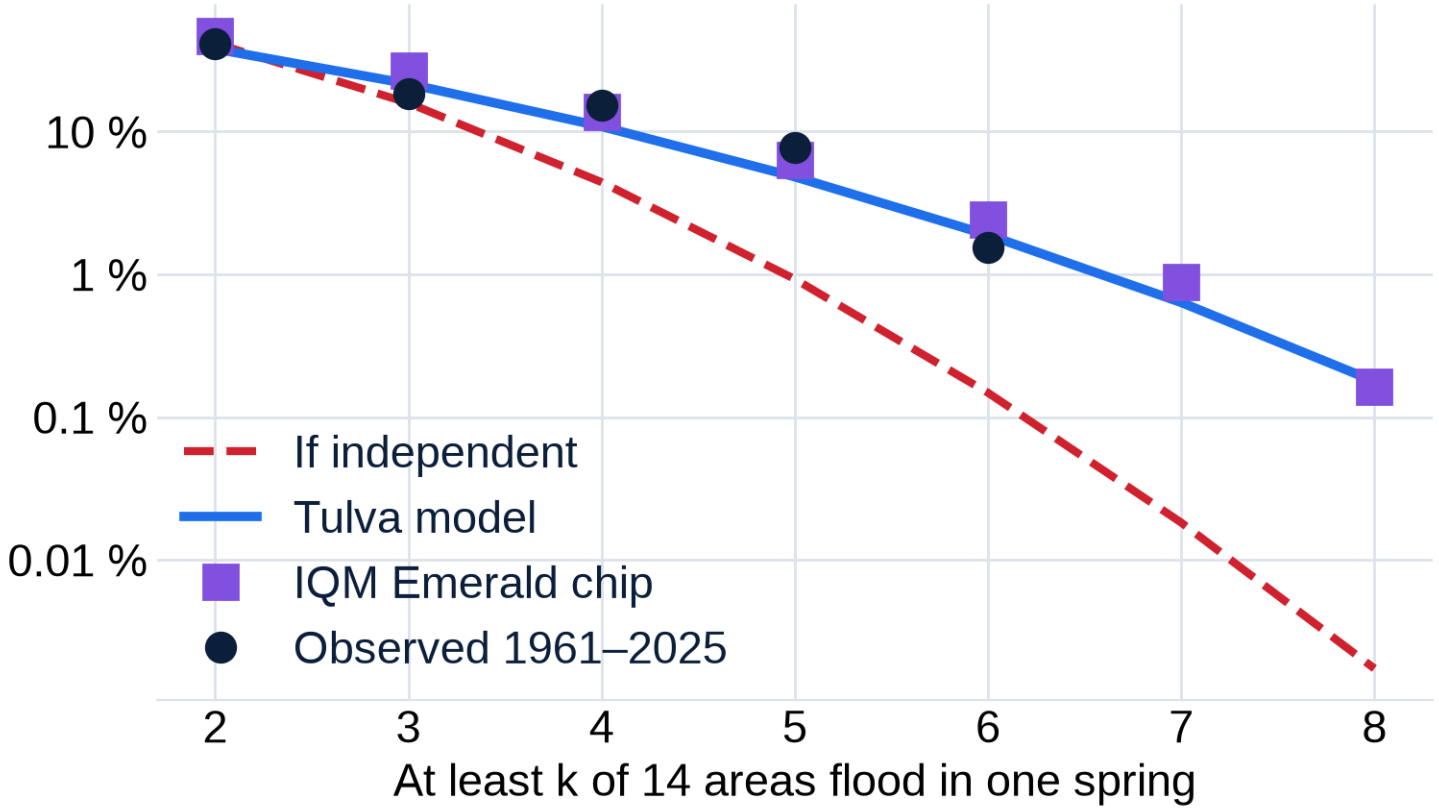
Source: SYKE 1961–2025, Tulva model (exact). Market price = 2.5–3.5× expected payout (AM Best, flood ILS)



Live app: joint-flood map → March snow → index price → quantum runs → 2005 replay

QUANTUM, IN PRACTICE

The full model fits on a chip today; amplitude estimation works on a 3-qubit piece of it.



At the 1-in-10 level the chip reproduces the joint tail.

ALL 14 AREAS ON IQM EMERALD

18 qubits

44 two-qubit gates

AMPLITUDE ESTIMATION

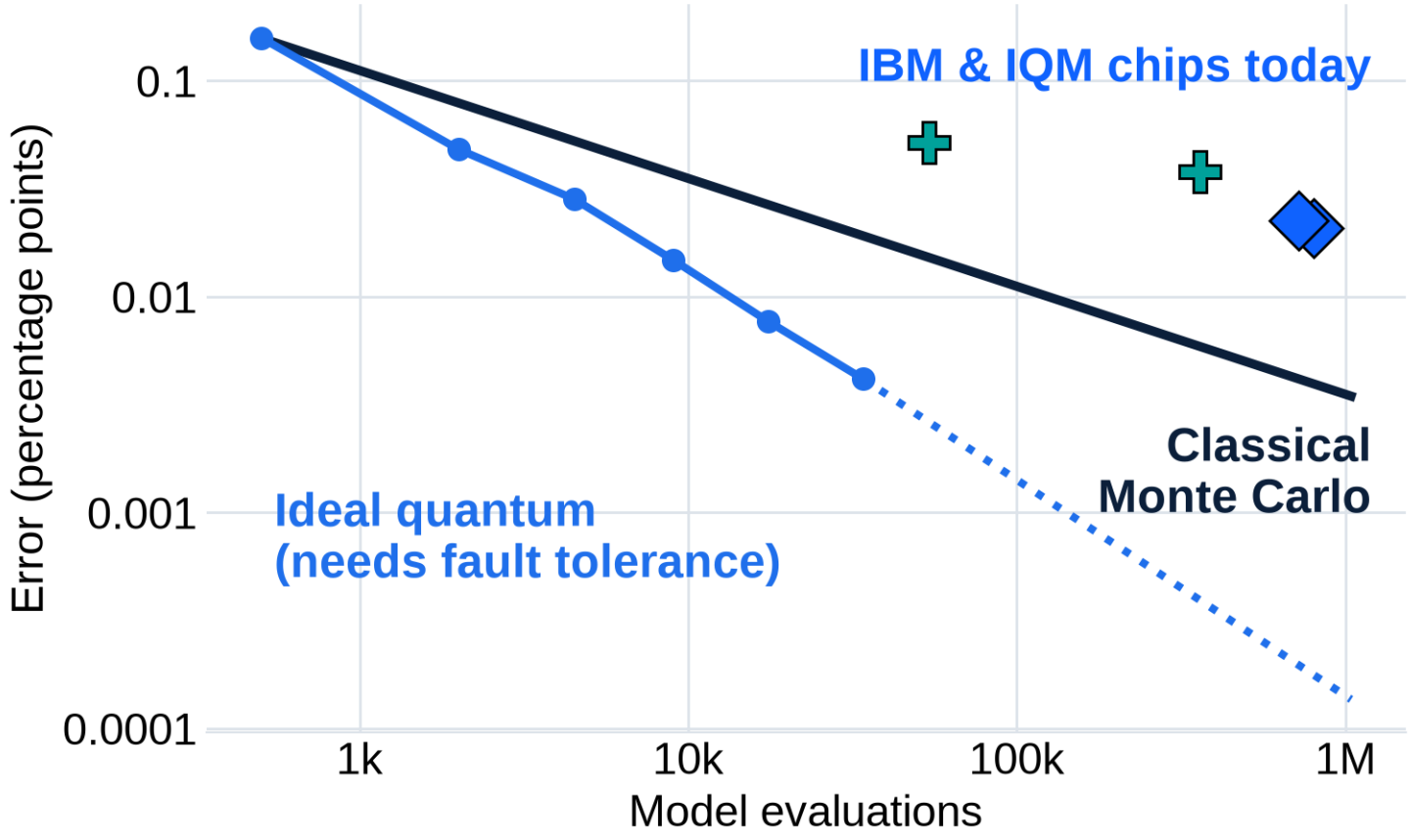
Kittilä and Ivalo both pass 1-in-50

Exact	**0.125 %**
IBM Marrakesh	**0.148 %**
IQM Garnet	**0.163 %**

Snow signal on the chip: snowy vs low March 3.1× (exact 2.7×)

IQM Emerald job 01a11cd4…, readout-corrected; IBM / IQM values calibrated (20 / 4 side-by-side copies); snow: Emerald, 10 copies

Classical Monte Carlo still wins today



Kittilä and Ivalo both pass 1-in-50: estimation error vs effort

Where quantum could pay off

Fault-tolerant hardware
~8k logical qubits needed

Expensive models
a hydrology model per scenario

More work per price
many perils, daily repricing

Error: QAE $\propto$ 1/N, Monte Carlo $\propto$ 1/√N

Source: Tulva simulations (qae-sim) + IBM Marrakesh / IQM Garnet runs; Chakrabarti et al., Quantum 5, 463 (2021)

BUSINESS

Who buys, why, and why now

WHO BUYS

Finnish non-life insurers

For the flood risk they keep. Four groups write 91 % of the market. Then reinsurers and ILS investors.

HOW WE EARN

Engine licence

€25k pilot season, then ~€75k per insurer per year (indicative). Pricing and trigger engine, licensed per insurer.

WHY NOW

30 Jan 2027

Finland gets a Solvency II flood capital factor: insurers need a joint-flood view now. The engine is quantum-ready as hardware matures.

SCALE

Finland → Nordics → Europe

Same engine, any snowmelt or river region. Nordic floods are costly (Gävle 2021: SEK 1.9bn). Storm Boris 2024: €2.15bn insured.

Ask **First step: a paid pilot with one Finnish insurer this autumn, before the 1 January reinsurance buy.**

Sources: Finance Finland 2024; Del. Reg. (EU) 2026/269; Finansinspektionen 2023 (Gävle); PERILS 2025 (Boris)

Tulva prices Finland's spring floods as a joint risk, and runs it on real quantum hardware.

BUILT THIS WEEKEND

SYKE data pipeline | Joint flood model | Index pricing | QAE on IBM & IQM

RESOURCES USED

SYKE open data | IBM Marrakesh | IQM Garnet & Emerald (Resonance) | Qrisp | Qiskit | AI coding agents

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack